

Data Structures

Fall 2025

General Overview

Data Structures



Algorithm
Analysis

How to organize information/data, usually in memory, so that it can be used efficiently?

How to analyze the performance of algorithms to evaluate their suitability for solving complex problems?

What is a Data Structure?

- In a general sense, any representation that is used for storing information is a data structure
 - Example: integers, structures, classes, arrays, etc.
- More typically, a **data structure** provides **a way for organizing a collection of data items**
 - Storing, accessing and modifying data items
- Most data structures have associated **algorithms to perform operations** on data
 - Search, insert, remove etc.

Costs and Benefits

- Each data structure requires
 - **Space** for storing data items
 - **Time** to perform each basic operation
 - Programming **efforts**
- **Cost** of a data structure
 - Time and space resources it consumes
- Choice of data structure depends on many factors
 - **Type** of data
 - **Frequency** with which various **data operations** are applied
- Hard to define data structure that performs better in all situations
 - Time and space tradeoff

Goals of This Course

- Learn the commonly used data structures
 - Form a programmer's basic data structure "toolkit"
- Case studies of data structures
- Examine the costs and benefits of every data structure

Motivational Example

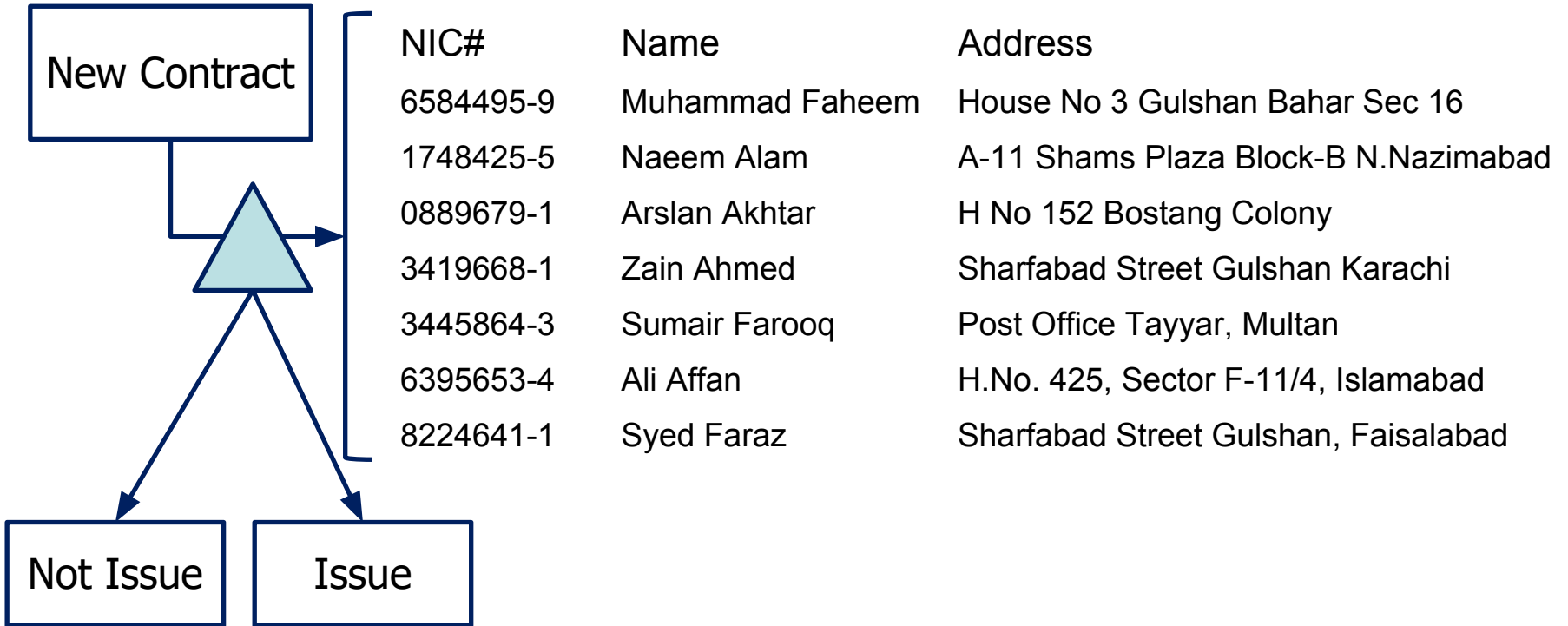
- A cellular service company provides contract to its 10 million users
- Due to new security enforcements, the company wants to **prevent issuing of multiple contracts to users**
- Method of Detecting Multiple Contracts
 - Before issuing a new contract to user
 - First search the **id of user** in existing contracts database
 - In case of failure ☐ issue a new contract
 - In case of success ☐ do not issue a new contract to user

Example: Linear Array Data Structure (1)

NIC#	Name	Address
6584495-9	Muhammad Faheem	House No 3 Gulshan Bahar Sec 16
1748425-5	Naeem Alam	A-11 Shams Plaza Block-B N.Nazimabad
0889679-1	Arslan Akhtar	H No 152 Bostang Colony
3419668-1	Zain Ahmed	Sharfabad Street Gulshan Karachi
3445864-3	Sumair Farooq	Post Office Tayyar, Multan
6395653-4	Ali Affan	H.No. 425, Sector F-11/4, Islamabad
8224641-1	Syed Faraz	Sharfabad Street Gulshan, Faisalabad

- Linear Array (with 10 million entries)
 - 3 arrays (NIC, Name, Address)
 - Structured array
 - Class's object array

Example: Linear Array Data Structure (2)



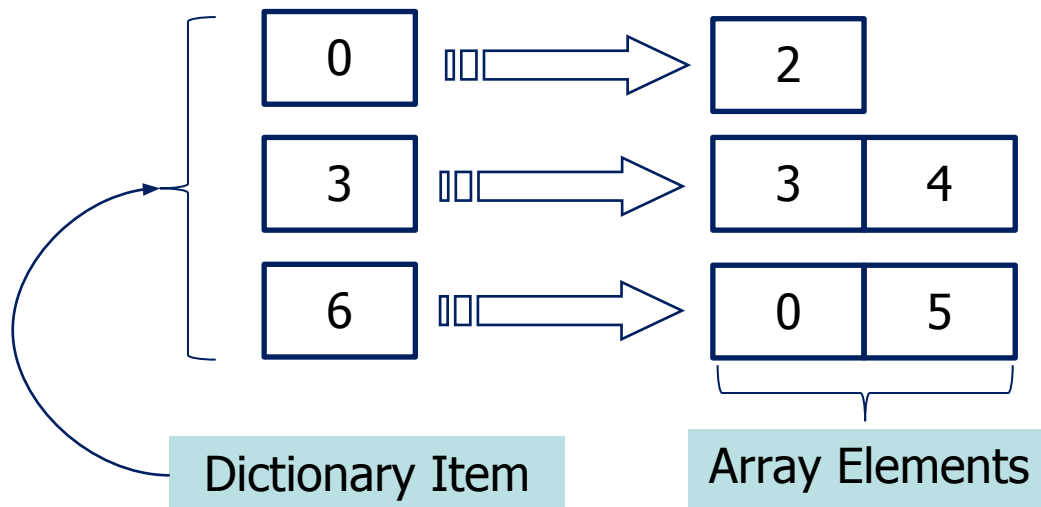
- Any disadvantage of linear array data structure?
- How to improve?

Example: Improved Data Structure

- Create a dictionary data structure
 - Group similar records together
 - Similarity in terms of first digit of NIC number
 - Add a dictionary entry for each distinct digit (0 – 9)
- Example: 3419668-1, 3445864-3, 1748425-5.
 - 3 and 1 are dictionary entries
- Existing contracts are searched in two steps
 1. Search the dictionary entry (i.e., group searching)
 2. Search contract within the group (i.e., with the same NIC digit)
- How much improvement w.r.t. linear array?

Example: Dictionary Data Structure (1)

NIC#	Name	Address
6584495-9	Muhammad Faheem	House No 3 Gulshan Bahar Sec 16
1748425-5	Naeem Alam	A-11 Shams Plaza Block-B N.Nazimabad
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Example: Dictionary Data Structure (2)

- Another possibility
 - Maintain pointers with structures (or records)
 - Non NULL pointer indicates presence of next record



Practice Task

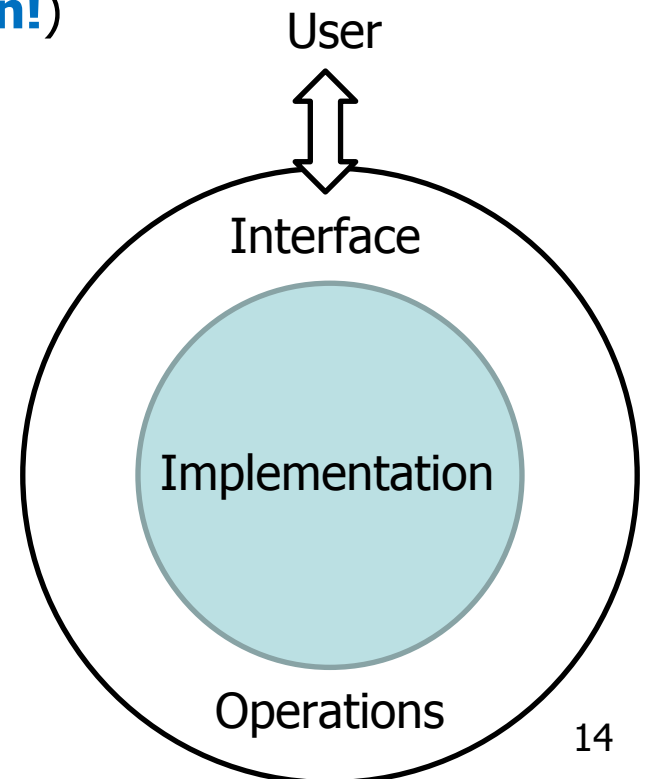
- Implement the example on the previous slides in C++ in two different programs. In the first, implement the data structure as a single array. In the second, implement the dictionary approach where the data is indexed by the first digit of the CNIC. In both programs, generate a random CNIC and search for it in your data structure. Do this searching operation in a loop of 500, to simulate 500 searches.
- Now, one by one, run both programs and note the running time. Use a Linux utility for checking the running time. Submit your programs and your result of running time for both (in a separate txt file).

What is Data Structure Efficiency?

- Two major measures of efficiency of a data structure or algorithm
 - Time and space
- Cost of data structure
 - Time and space resources it consumes
- Efficiency of a data structure depends on its operations
 - Makes a difference between program running in few seconds or days
- Difficult to define data structure that performs better in all situations

Abstract Data Types (1)

- A definition of data type solely in terms of
 - Set of related data items (or values)
 - Set of operations on the data
- Separation of logical properties from the implementation details
 - Hide implementation details (**Encapsulation!**)
- **What** not **how** is the focus!



Data Structures vs. ADT

- A data structure is physical implementation of an ADT
 - Each operation associated with ADT is implemented by one or more subroutines in the implementation
- **Data structure** usually refers to an organization of data in main memory
- Example: Stack, or Queue are ADTs, but a C++ Queue implemented in a particular class with particular functions is a concrete data structure.
- The relationship between ADT and Data Structure is a bit like the relationship between an interface (class.h file – tells WHAT is done) and a class implementation (the class.cpp file – tells HOW it is done).

Array as an ADT

- An array is defined as
 - Ordered collection of a fixed number of elements
 - All elements are of the same data type
- Basic operations
 - Direct access to each element in the array
 - Values can be retrieved or stored in each element

C/C++ Implementation of an Array ADT

```
datatype arrayName[intExp];
```

As an ADT

Ordered

Fixed Size

Homogeneous

Direct Access

In C/C++

Index: 0,1,2, ... SIZE-1

intExp is constant

dataType is the type of all elements

Array subscripting operator []

Summary

What is a data structure?

- In a general sense, any representation that is used for storing information is a data structure
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Where is a data structure helpful?

- Data structure can make the difference between a program running in a few seconds or many days
 - Efficiency of data structure
- **What is an ADT?**
 - A logical representation of a data structure.

Any Question So Far?

